Mangler så bare $|\mathcal{M}_{fi}|^2$. I CoM dør $\mathcal{M}_0$ da vi får noget på formen $\varepsilon^\mu_{out} P_\nu P'_\mu \varepsilon^\nu_{in}$ og $\varepsilon^\mu_{out} k'_\mu = 0 = \varepsilon^\mu_{out} P'_\mu$ da p & k er antiparallelle. Kan måske bare finde $|\mathcal{M}_{fi}|^2$ i rest frame da den er frame-independent? Vil så trivially være kun seagull, altså

$$|\mathcal{M}_{fi}|^2 = |\mathcal{M}_0|^2 = 4e^4 \left(\varepsilon^\mu_{in} \varepsilon^\nu_{out} \eta_{\mu\nu}\right)\left(\varepsilon^\sigma_{in} \varepsilon^\delta_{out} \eta_{\sigma\delta}\right)$$

We take average over polarizations so

$$|\mathcal{M}_{fi}|^2 = 4e^4 \frac{1}{2}\left(\sum_\lambda \varepsilon^\mu_{in} \varepsilon^\sigma_{in}\right)\left(\sum_\lambda \varepsilon^\nu_{out} \varepsilon^\delta_{out}\right)\eta_{\mu\nu}\eta_{\sigma\delta}$$

$$\overset{6.31}{=} 2e^4\left(\delta^{\mu\sigma} - \frac{k^\mu k^\sigma}{|\vec{k}|^2}\right)\left(\delta^{\nu\delta} - \frac{k'^\nu k'^\delta}{|\vec{k}'|^2}\right)\eta_{\mu\nu}\eta_{\sigma\delta}$$

$$= 2e^4\left(\delta^\sigma_\nu \delta^\nu_\sigma - \delta^\sigma_\nu \frac{k'^\nu k'_\sigma}{|\vec{k}'|^2} - \delta^\nu_\sigma \frac{k_\nu k^\sigma}{|\vec{k}|^2} + \frac{k_\nu k^\sigma k'^\nu k'_\sigma}{|\vec{k}|^2 |\vec{k}'|^2}\right)$$

$$= 2e^4\left(\delta^\sigma_\sigma - \frac{k'^\sigma k'_\sigma}{|\vec{k}'|^2} - \frac{k_\sigma k^\sigma}{|\vec{k}|^2} + \frac{k_\nu k'^\nu k^\sigma k'_\sigma}{|\vec{k}|^2 |\vec{k}'|^2}\right)$$

Ved ikke om men tidligere kun brugte $\varepsilon^0 = 0$ så sum bliver over i,j i stedet så

$$|\mathcal{M}_{fi}|^2 = 2e^4\left(\delta^i_i - \frac{k'^i k'_i}{|\vec{k}'|^2} - \frac{k^i k_i}{|\vec{k}|^2} + \frac{k_i k'^i k^j k'_j}{|\vec{k}|^2 |\vec{k}'|^2}\right)$$

$$= 2e^4\left(3 - 1 - 1 + \frac{(\vec{k}\cdot\vec{k}')^2}{|\vec{k}|^2 |\vec{k}'|^2}\right) = 2e^4\left(1 + \cos^2(\theta)\right)$$

Så

$$\frac{d\sigma}{d\Omega} = \frac{1}{64\pi^2}\frac{1}{E_I^2} 2e^4\left(1+\cos^2(\theta)\right)$$

$$= \frac{1}{32\pi^2}\frac{1}{E_I^2} e^4\left(1+\cos^2(\theta)\right)$$